

JADY PAMELLA

PhD Fellow in Digital Twins, Aarhus University | AI Engineer focused on AI Safety | Peer-Reviewed in Elsevier

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SUMMARY

PhD Fellow in digital twins at Aarhus University and AI Engineer focused on AI safety, computer vision, and risk management for autonomous systems. My doctoral research builds digital twins that verify district cooling installations run as they were designed (SWiM, funded by the Grundfos Foundation), at the intersection of AI safety, cybersecurity, and critical-infrastructure resilience. Built a real-time LiDAR perception pipeline for safe drone landing below forest canopy at Deep Forestry, reaching F1 0.926 at 53 ms/frame on NVIDIA Jetson Orin NX, with a 29× end-to-end latency reduction (5.8 s to 200 ms) (MSc thesis defended 2026 at Stockholm University, Swedish Institute Scholar). Second master's in Cybersecurity with concentration in Risk Management from University of Brasilia (2026), co-author of a peer-reviewed Elsevier Computers and Security paper (IF 5.6). 12+ years of regulated public-sector engineering at Bank of Brasilia. First place in three hackathons.

Research focus: Safety-critical autonomous perception | Point-cloud deep learning (PointNet/++/DGCNN) | Inference-latency optimization on edge | Trustworthy and verifiable AI | Cybersecurity for safety-critical and public-sector systems

PEER-REVIEWED PUBLICATIONS

Enhancing Cybersecurity in the Judiciary: Integrating Additional Controls into the CIS Framework — 2025

- Computers and Security (Elsevier, Impact Factor 5.6). Co-authored peer-reviewed study proposing an expanded CIS cybersecurity model for digital protection across the Brazilian judiciary. DOI 10.1016/j.cose.2025.104584.

Cybersecurity Protection in the Brazilian Judiciary: A Comparative Analysis of State Court Security Structures — 2024

- NAVUS, Journal of Technology and Management, v. 14, pp. 1–22. Research analyzing cybersecurity maturity and structural gaps in state courts across Brazil. DOI 10.22279/navus.v14.2032.

Cybersecurity Challenges: An Analysis of the Brazilian Public Administration — 2026

- Revista Tecnologia e Sociedade (UTFPR), v. 22, n. 69, p. 157–178. Peer-reviewed study identifying the main cybersecurity governance challenges faced by the Brazilian public administration, from the perspective of managers and specialists. Read the article. DOI 10.3895/rts.v22n69.20157.

Critical Infrastructure Under Threat: Business Risks for Cybersecurity in Metropolitan Transport Systems — 2025

- 36th ENANGRAD, Fortaleza, Brazil. Peer-reviewed conference paper on operational and governance exposures in metropolitan transport systems and the controls needed to mitigate them. Proceedings entry.

DATASETS AND THESES

Security Controls for Judiciary Risk Management — 2025

- IEEE DataPort. Open dataset of the security controls assessed in the judiciary study, published so the analysis behind the Computers and Security paper can be reproduced. DOI 10.21227/h9q6-sv58.

Automated Detection of Safe Landing Locations in Below-Canopy Forest Conditions — 2026

- MSc Thesis, Department of Computer and Systems Sciences (DSV), Stockholm University, in collaboration with Deep Forestry AB. Supervisor: Prof. Rahim Rahmani. Real-time LiDAR perception for autonomous landing under the forest canopy.

An Information Security Control Baseline to Mitigate Business Risk in the Sanitation Sector — 2026

- Professional Master's Dissertation, PPEE, University of Brasilia, record PPEE.DM-124/2026. Supervisor: Prof. Rafael Rabelo Nunes. A baseline of information security controls for water and sanitation utilities, derived from the business risks of the sector.

RESEARCH AND INDUSTRY EXPERIENCE

Aarhus University – PhD Fellow, Digital Twins for District Cooling (SWiM) — Aug 2026 - Present (Aarhus, Denmark)

- Verifying that district cooling installations are commissioned as they were designed, combining Building Information Modelling, AI and digital twins. Part of SWiM, a five-year project on sustainable water-based cooling for megacities funded by the Grundfos Foundation, supervised by Peter Gorm Larsen and Carl Schultz.
- Work at the intersection of AI safety, cybersecurity, and critical-infrastructure resilience.

Deep Forestry AB – AI Research Engineer (MSc thesis collaboration with Stockholm University) Dec 2025 - Jun 2026 (Uppsala, Sweden)

- Designed and evaluated a hybrid geometric-neural perception pipeline (RANSAC + PointNet, PointNet++, DGCNN) for real-time autonomous landing-zone detection below the forest canopy from LiDAR point clouds, reaching F1 0.926 on a held-out flight session.
- Reduced end-to-end inference latency 29× (5.8s to 200ms, 53 ms/frame) on NVIDIA Jetson Orin NX through KD-Tree indexing and voxel downsampling, evaluated under Group K-Fold session-level cross-validation on real flight data.

LabRisk, Cybersecurity and Risk Lab, University of Brasilia (UnB) – Cybersecurity Research Engineer Aug 2024 - Present (Brasilia, Brazil)

- Research on cybersecurity management, risk, and privacy for the public sector (CIS controls, GDPR-aligned frameworks, judiciary cybersecurity in Brazil). Four peer-reviewed publications and one open dataset listed above.

Stockholm University and Drivhuset Stockholm – Teaching Assistant (IoT) and Student Ambassador 2024 - 2026 (Stockholm)

- Supported 200+ students in IoT coursework (Python, MQTT, edge computing). Co-produced student hackathons and entrepreneurship events at Stockholm University through Drivhuset.

SoiQet (Social IQ Set) – Founder and CEO Aug 2025 - Present (Aarhus, Denmark)

- Co-founded an LLM-powered multi-tenant SaaS on Azure with in-house retrieval-augmented generation, structured outputs, fine-tuned LLM, OAuth across six networks, Stripe billing, and Supabase row-level security. Building cross-provider model evaluation to compare cost and quality across OpenAI, Anthropic, and others.

Bank of Brasilia (BRB) – IT Analyst and DevOps Engineer / IT Manager Dec 2014 - Present (Brasilia, Brazil)

- Production NLP chatbot (Python, TensorFlow) cutting customer response time 40% for 100K+ users during a Silicon Valley engagement. GDPR-compliant customer-data analytics for the instant payments system (PIX) on PySpark. CI/CD on Jenkins, GitLab, Docker, Kubernetes, OpenShift. Deployment time cut 60%.

EDUCATION

Stockholm University - MSc Computer and Systems Sciences (AI and IoT), Swedish Institute Scholar (SISGP) Aug 2024 - May 2026 (Stockholm)

University of Brasilia (UnB) - MEng Cybersecurity, peer-reviewed in Elsevier Mar 2024 - May 2026 (Brasilia)

UPIS Faculdades Integradas - BSc Information Systems 2010 - 2013 (Brasilia)

AWARDS, CERTIFICATIONS, AND SKILLS

Speaking and Judging: Speaker, "New Tech, Old Rules" panel, Nordic Tech Week, Stockholm (2026). Judge, Apart Research Global South AI Safety Hackathon, Technical AI Safety, AI Governance and AI Security tracks (2026).

Awards: 1st place EpiMinds Google AI Hackathon (2026), Kong AI-Assisted Coding (2025), Kolomolo AI-Assisted Coding (2025). Swedish Institute Scholarship for Global Professionals (2024). Impact Cup, Drivhuset (2025). Strategic Configuration Management, Efinance (2019).

Certifications: EA Sweden Fellowship in Effective Altruism (2026), Designing and Building AI Products (MIT, 2022), EXIN DevOps Master (2020), Lean IT Foundation (2019), COBIT 5 (ISACA, 2015), ITIL V3 (2015), TOEFL iBT 101/120 (ETS, 2024).

Research and ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face, PointNet/++/DGCNN, RANSAC, KD-Tree, voxel downsampling, LLM fine-tuning, RAG, NLP, MLOps, model deployment, evaluation harnesses.

Programming and Platform: Python, SQL, TypeScript, JavaScript, Bash. Azure, GCP, AWS, Kubernetes, Docker, Spark, Airflow, Kafka, PostgreSQL, MongoDB.

Cybersecurity and Governance: CIS Controls v8, GDPR, Risk Management, Audit and Compliance, COBIT, ITIL, Preliminary Hazard Analysis (PHA), Bow Tie.

Languages: Portuguese (native), English (fluent, TOEFL 101/120), Spanish (intermediate), Swedish (SFI Level C), Danish (beginner, learning).